Casey Barton

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PROFILE

I build the evaluation harness before I build the feature — for non-deterministic systems, 'it looked good in testing' is not an answer. I define what working means, instrument it, and let the evals drive the design. Fail-closed by default: guardrails live in the orchestration layer, and the defined 'what happens when it's unsure' path is halt-and-escalate, never guess.

Three years as a founder running an evidence-grade agent control plane from Honolulu: hand-rolled multi-agent loops with function calling, structured outputs, typed handoff contracts, retrieval over my own estate, and 65 MCP connectors — embedded-style delivery, measured against written outcomes before deployment.

CORE COMPETENCIES

Engineering: Python · TypeScript · Node.js · SQL · Bash · REST APIs · webhooks · containers · GitHub Actions (CI/CD) · Supabase · Postgres

AI & Agents: LLM evaluation harnesses · replay loops · guardrails · failure-mode analysis · hand-rolled multi-agent loops · function calling · structured outputs · structured extraction · RAG · prompt & context engineering · MCP protocol · human-in-the-loop control

Deployment: embedded delivery (scope with operators, prove end to end, hand over something they can run) · demos & POCs · monitoring & observability · latency & token-cost benchmarks · verification gates · audit receipts · idempotent automation · evidence boundaries

EXPERIENCE

Founder / Applied AI Systems Architect 2023 – Present

GlacierEQ — Honolulu, HI

- Build the evaluation harness before the feature: prompt and configuration changes validated by eval harnesses with replay loops before anything touches production — I define what working means, instrument it, and let the evals drive the design.
- Reason about failure modes explicitly: guardrails in the orchestration layer — typed state transitions, approval gates on consequential actions, fail-closed recovery; 'cannot determine' is a first-class answer, not a guess.
- Ship structured extraction and retrieval pipelines across a **1,395-repository** estate — structured facts out of messy documents, deterministic checks gating sensitive operations.
- Ship production GenAI systems with structured outputs, function calling with typed tool schemas, and retrieval-augmented generation, backed by a **989-test** control plane — **62/62** verification checks green.
- Integrate **65 MCP connectors** plus APIs, webhooks, and data pipelines into environments I don't own — evidence boundaries and audit receipts built for client-cloud compliance regimes.
- Own outcomes against a written measurement story: KPIs, baselines, and instrumentation agreed before deployment — then deliver against them.

EDUCATION

University of Hawaii at Manoa 2016
Bachelor of Science

AWS Cloud Institute 2025
Cloud Computing program

Colorado State University
Undergraduate coursework

SELECTED PROOF

Job-App Helix — attributable campaign engine

- Composes assessments into campaign-level **GO / NO-GO** with immutable evidence models, stable finding codes, and fail-closed decisions.
- Recovery restricted to contingency evidence declared before reassessment — the engine never silently tunes values until they pass.

GlacierEQ estate — agent systems portfolio

- Durable patterns for claim tracking, evidence boundaries, audit receipts, and human-in-the-loop control across the full estate.
- Connects multiple technical surfaces — agents, APIs, databases — into one coherent, verifiable working system.

SKILLS

Languages: Python · TypeScript · SQL · Bash · JavaScript / Node.js

AI / Agents: LLM evaluation harnesses · replay loops · guardrails · hand-rolled multi-agent loops · function calling · structured outputs · structured extraction · RAG · MCP protocol · typed handoff contracts · prompt & context engineering

Delivery: embedded delivery · demos & POCs · monitoring & observability · latency & token-cost benchmarks · verification gates · audit receipts · idempotent automation

Evidence boundary: portfolio repositories are independent work. Domain-inspired demonstrations are not claims of company affiliation, production deployment, or certification.